Bayesian Inference with a Binomial Model

Felipe Sanchez

March, 2026

To illustrate how Bayesian inference works in practice, we implement a binomial model that aims to estimate the probability of a baby being born as a girl in a given hospital. Specifically, we want to estimate the probability that a newborn baby is a girl.

We assume (and this is a reasonable assumption) that the outcome variable Y, which takes the value 0 (boy) or 1 (girl), is generated by Bernoulli trials with some unknown parameter θ, representing the probability that a baby is a girl. The probability of observing k girls out of n births given θ is

$$P(\text{girls} = k \mid n, \theta),$$

which is well described by a binomial model.

One advantage of this model is that, if we assume a Beta prior distribution for θ, we can derive the posterior distribution analytically.

By the way, what is the **frequentist estimator** of θ?

Prior

$$\theta \sim \text{Beta}(a, b)$$

with expectation

$$E(\theta) = \frac{a}{a+b}.$$

Posterior

$$\theta \mid k, n \sim \text{Beta}(a + k,\, b + n - k)$$

with expectation

$$E(\theta \mid k, n) = \frac{a+k}{a+b+n}.$$

Note: When the posterior distribution has the same functional form as the prior distribution, the prior is called *conjugate*.

Simulated Data

Based on demographic knowledge, we simulate data for the first thousand babies born in a hospital since the beginning of 2017 (1 denotes a girl and 0 denotes a boy).

```
set.seed(202)
girl = rbinom(n = 1000, size = 1, prob = 0.4854369)
```

Exercises

1. Compute and plot the mean for the first 20 observations, then 50, then 100, and so on.
2. Plot the data to examine the distribution of the simulated outcomes for different sample sizes.
3. What is the frequentist estimator of θ for each sample size?

1.3 Plotting Posterior Distributions

(a) What is the posterior distribution of the parameter θ if we use conjugate priors?

(b) What are the posterior distributions given the prior distributions that we proposed earlier?

(c) Plot the prior, likelihood, and posterior distributions using the following code.

```
a = 5; b = 5
th = seq(0, 1, by = 0.001)

k = sum(girl[1:20])   # sample size: 20 observations
n = length(girl[1:20])

# Beta(5,5) prior
plot(th, dbeta(th, a, b), ylim = c(0,5), col = "darkgrey", t = "l")

# Likelihood for 20 observations
lines(th, dbeta(th, k+1, n-k+1), col = "red", t = "l")

# Posterior distribution
lines(th, dbeta(th, a+k, b+n-k), col = "black", lwd = 3, t = "l")
```

(d) Now keep the same prior, but increase the sample size to $n = 100$.

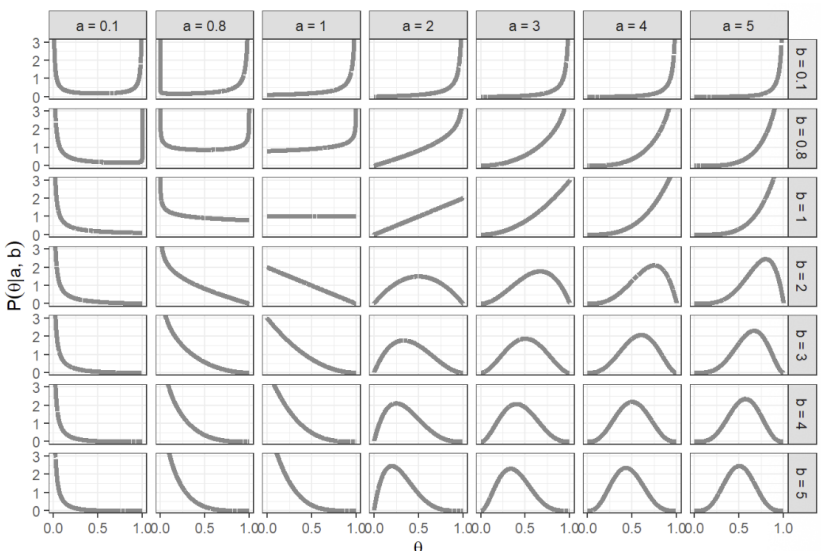

Figure 1: Beta distribution

(e) Plot the prior, likelihood, and posterior for an informative prior where

$$a = 1000, \qquad b = 1000 \times \frac{1 - 0.4854369}{0.4854369}.$$

Below are helper plots for the priors.

```
# Beta(0.5, 0.5)
plot(th, dbeta(th, 0.5, 0.5), ylim = c(0,30), col = "darkgrey", t = "l")

# Informative prior with expectation 0.4854369
plot(th, dbeta(th, 1000, 1000*(1-0.4854369)/0.4854369),
     ylim = c(0,400), col = "darkgrey", t = "l")
```

(f) How does our inference change when we increase the sample size?

(g) Repeat the analysis using the prior

$$a = 0.5, \qquad b = 0.5.$$